COSC300 GROUP 17 SDLC

GROUP MEMBERS :

RAKGETSE PI 202204262

KEKANA PL 240064443

NETSHANDAMA PL 202225805

MOGORU PM 202105210

MUTHALA PW 202343544

LUSUNZI R 240021616

MASHABNA NB 201820548

SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC) METHODOLOGY REPORT

1. INTRODUCTION

Taxila Secondary School currently faces challenges with fragmented communication, high printing costs, and limited after-hours academic support for students. The Taxila Learning Hub aims to digitize the learning environment by centralizing notes, grades, and assignments while providing an integrated AI-driven "Tutor Expert" to bridge the gap between classroom teaching and home study. Taxila Secondary School currently operates using a fragmented communication and resource distribution model. While WhatsApp groups are used for quick updates, academic materials (notes, past papers, and worksheets) are primarily distributed via physical handouts or scattered digital messages.

This leads to the following critical issues:

Information Loss: Important study materials get buried in chat histories or are physically lost by students, leading to gaps in exam preparation.

High Operational Costs: The school spends a significant portion of its budget on paper and toner for printing thousands of handouts weekly.

The 'After-Hours' Gap: Students who struggle with homework in the evenings have no way to get immediate academic guidance, often leading to incomplete assignments and frustration.

Lack of Analytics: Teachers have no way to track which students have actually opened the study material or where the most common learning "stumbling blocks" are occurring."

2. **Setting Goals**: What "Success" Looks Like

Success for the Taxila Learning Hub is defined by achieving the following SMART (Specific, Measurable, Achievable, Relevant, Time-bound)

Goal 1 (Resource Access): To ensure 100% of curriculum notes for Grades 8–12 are digitized and accessible for offline download by the start of the next academic term.

Goal 2 (Tutoring Support): To integrate a Tutor Expert AI capable of answering 90% of CAPS-aligned curriculum queries instantly, providing 24/7 support to students off-campus.

Goal 3 (Cost Reduction): To reduce the school's academic printing expenditure by at least 35%

Goal 4 (Engagement): To achieve a 80% student adoption rate, where students use the app at least three times a week for revision or assignment tracking

Technical Feasibility

The selection of Google Firebase as the backend infrastructure ensures that the Taxila Learning Hub remains operational 24/7. By offloading processing to the cloud, we bypass local power instability (load shedding) and provide 'Offline Persistence,' which allows students to study without an active internet connection, directly addressing the socio-economic challenges of the student body

Legal & Schedule Feasibility (The "Law & Time" Check)

POPIA Compliance: In South Africa, protecting student data is a legal requirement. The app will use End-to-End Encryption for grades. No personal student data (like ID numbers) will be stored in plain text or shared with third parties.

Operational Feasibility (The "People" Check)

An app is only successful if the teachers and students actually use it.

Teacher Workload: To ensure teachers aren't overwhelmed, we will include a "One-Click Upload" feature where they can sync their existing Google Drive or WhatsApp folders directly to the app.

Student Engagement: The Tutor Expert acts as a "hook." Students who might ignore a PDF will use the app to get instant help with a difficult Math problem at home.

Based on the technical, economic, and operational investigations, the Taxila Learning Hub is highly feasible. The cost-savings from paper alone provide a sustainable financial model, and the use of cloud-based AI bypasses the hardware limitations of student devices.

Economic Feasibility (The "Can we afford it?")

We conducted a Cost-Benefit Analysis:

The Cost: Hosting the database and AI API calls is estimated at R600–R1,000 per month for the whole school.

The Saving: Currently, Taxila spends approximately R4,500 per month on paper, toner, and photocopier maintenance for handouts.

The Verdict: The project is Economically Viable. By moving just 50% of handouts to the digital Hub, the school saves enough money to pay for the app's running costs four times over based on the technical, economic, and operational investigations.

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2. Planning Phase

The planning phase is the first stage of the Software Development Life Cycle. During this phase, the main goal is to identify the problem that the system will solve and determine whether the proposed solution is feasible.

For the School Digital Platform project, the planning phase focuses on identifying the challenges faced by schools when managing communication and monitoring student academic performance. Many schools rely on manual communication methods such as printed newsletters or disconnected digital tools. This makes it difficult for parents and students to receive timely updates and for teachers to monitor student progress effectively.

During the planning phase, the project objectives are defined and the overall scope of the system is determined. The main objective of the School Digital Platform is to centralize academic information, announcements, and communication into a single digital system that is accessible through both web and mobile platforms.

A feasibility study is also conducted to determine whether the project is technically and operationally possible. This includes evaluating the required technologies, system resources, and development tools. The planning phase also identifies the key stakeholders of the system, including students, teachers, parents, and school administrators.

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3. Analysis Phase

The analysis phase focuses on understanding the system requirements and identifying the features that the system must provide. During this stage, developers gather detailed information about how the system will be used and what functionalities it must support.

For the School Digital Platform project, the analysis phase involves identifying the needs of the different users who will interact with the system. Students will require access to homework tasks, academic announcements, and learning materials. Teachers will need tools to monitor student performance and receive alerts when students are at risk academically. Parents will require access to student progress reports and school announcements.

The analysis phase also involves studying existing systems and identifying their limitations. Many current systems do not integrate student performance data with communication tools, which makes it difficult for teachers to detect students who may be struggling academically.

The results of the analysis phase are documented in the **Software Requirements Specification (SRS)** document. This document describes both functional requirements (what the system should do) and non-functional requirements (performance, security, usability, etc.).

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4. System Design Phase

The School Digital Platform will follow a three-layer system architecture consisting of the frontend, backend, and database.

4.1. Frontend (Presentation Layer)

The frontend is the part of the system that users interact with. It includes the school website and the mobile application.

Users such as students, parents, teachers, and administrators will be able to:

Log into the system

View announcements and newsletters

Access homework and academic information

Receive notifications and updates

4.2. Backend (Application Layer)

The backend handles the main system logic and connects the frontend to the database.

The backend will manage:

User authentication and login

Academic data processing

Announcement management

AI model integration

Notification services

Both the website and mobile application will communicate with the backend through APIs.

4.3. Database (Data Layer)

The database stores all system information, including:

User accounts

Student academic records

Attendance data

Homework and assignments

School announcements

A centralized database ensures that both the website and mobile app access the same information.

4.4. AI Module

The system will include an Artificial Intelligence module that analyses student data such as marks, attendance, and homework completion.

The AI model will classify students into:

Low Risk

Medium Risk

High Risk

Teachers will receive alerts when a student is identified as High Risk, allowing early intervention.

The system will also include AI content summarization for announcements and an AI Tutor Expert to help students understand difficult topics.

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5. Implementation Phase

The implementation phase is the stage where the system is developed based on the design created during the previous phase. In this phase, developers write the system code,

integrate the different system components, and perform testing to ensure that the system functions correctly.

For the School Digital Platform project, the implementation phase will involve developing several components including the backend server, website interface, mobile application, and the Artificial Intelligence module.

The system will be developed using modern programming languages and technologies commonly used in web and mobile development. The **frontend interface** will be developed using **HTML, CSS, and JavaScript** to create a responsive and user-friendly website. The **mobile application** may be developed using cross-platform technologies such as **Flutter or React Native**, allowing the system to run on both Android and iOS devices.

The **backend system** will be implemented using server-side technologies such as **Node.js or Python**, which will handle user authentication, data processing, and communication between the website, mobile application, and the database. The system database may use technologies such as **MySQL or PostgreSQL** to securely store academic records, attendance information, and user data.

The **Artificial Intelligence component** will be implemented using **Python and machine learning libraries such as Scikit-learn**. The AI model will use a **Logistic Regression algorithm** to analyse historical student data including marks, attendance, and homework completion rates. Based on this data, the system will predict whether a student is likely to pass or fail before final examinations and classify students into risk categories.

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6. Maintenance Phase

The maintenance phase occurs after the system has been deployed and is being used by its users. During this phase, the system is monitored and improved to ensure that it continues to function effectively.

Maintenance activities include fixing software bugs, improving system performance, updating system features, and ensuring that the platform remains secure. As more student data becomes available, the Artificial Intelligence model may also be retrained or updated to improve prediction accuracy.

For the School Digital Platform, maintenance will ensure that the system remains reliable and continues to meet the needs of schools, teachers, students, and parents.

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7. Conclusion

The Software Development Life Cycle provides a structured and organized approach to developing software systems. By following the phases of planning, analysis, design,

implementation, and maintenance, development teams can ensure that software projects are completed successfully and meet the needs of users.

In this project, the SDLC methodology will guide the development of the School Digital Platform. The structured approach provided by SDLC will ensure that the system is carefully planned, properly designed, effectively implemented, and continuously maintained to support communication and academic monitoring in schools.